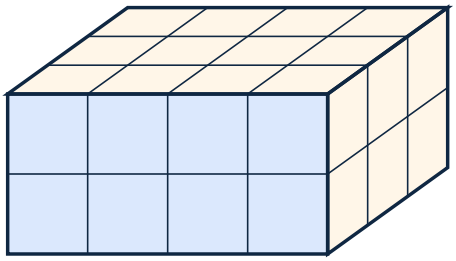
Volume

How many unit cubes fill the box? Count one layer, then the layers.

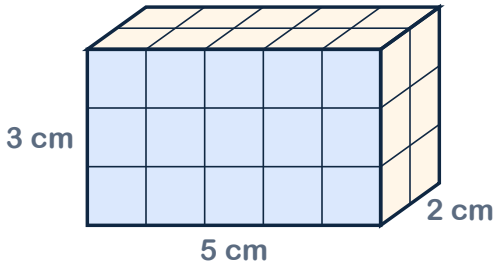
A box 4 cubes long, 3 cubes wide, 2 cubes tall



One layer: 4 rows of 3 = 12 cubes.
2 layers: 12 + 12 = 24 cubes.
Volume is how many cubes fill it.

$4 \times 3 \times 2 = 24$ cubes

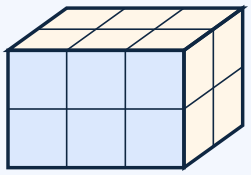
Length × width × height



Each cube is 1 cm on every edge: a cubic cm.
Base: 5 × 2 = 10 squares. Height: 3 layers.

$5 \times 2 \times 3 = 30$ cubic cm

Your turn!



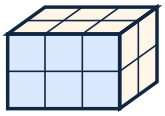
volume = ☐ cubes

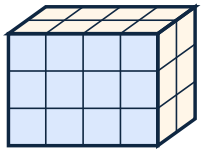
Volume: Practice

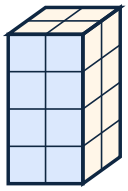
One layer, then the layers. Length × width × height.

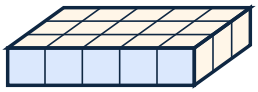
Part 1 Count the cubes

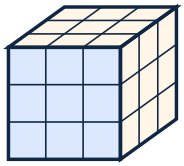
Some cubes are hidden. Count one layer, then the layers.

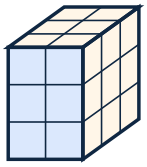
1)  ☐ cubes

2)  ☐ cubes

3)  ☐ cubes

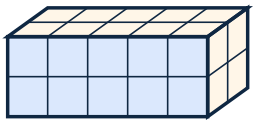
4)  ☐ cubes

5)  ☐ cubes

6)  ☐ cubes

Part 2 Layers

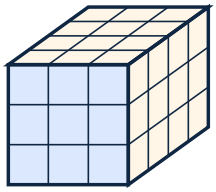
Cubes in one layer, then how many layers.

7) 

one layer: 5 × 2 = ☐

layers: ☐

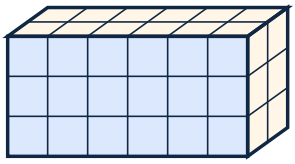
10 × 2 = ☐ cubes

8) 

one layer: 3 × 4 = ☐

layers: ☐

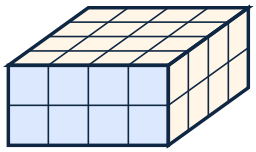
12 × 3 = ☐ cubes

9) 

one layer: 6 × 2 = ☐

layers: ☐

12 × 3 = ☐ cubes

10) 

one layer: 4 × 4 = ☐

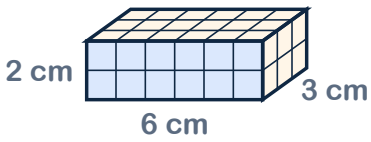
layers: ☐

16 × 2 = ☐ cubes

Volume: Practice (continued)

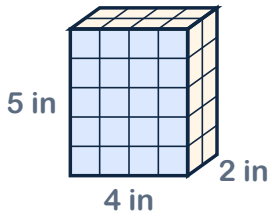
Part 3 Length × width × height Use the numbers on the edges.

11)



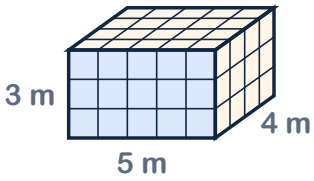
$6 \times 3 \times 2 =$ ☐ cubic cm

12)



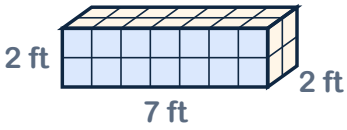
$4 \times 2 \times 5 =$ ☐ cubic in

13)



$5 \times 4 \times 3 =$ ☐ cubic m

14)



$7 \times 2 \times 2 =$ ☐ cubic ft

Part 4 Find the volume No help this time.

15) $8 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm} =$ ☐ cubic cm

16) $5 \text{ in} \times 5 \text{ in} \times 4 \text{ in} =$ ☐ cubic in

17) $9 \text{ m} \times 2 \text{ m} \times 3 \text{ m} =$ ☐ cubic m

18) $6 \text{ cm} \times 4 \text{ cm} \times 5 \text{ cm} =$ ☐ cubic cm

19) $10 \text{ ft} \times 3 \text{ ft} \times 2 \text{ ft} =$ ☐ cubic ft

20) $7 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm} =$ ☐ cubic cm

Volume: Practice (continued)

Part 4 Find the volume (continued)

21) 4 in × 4 in × 4 in = ☐ cubic in

22) 12 m × 2 m × 3 m = ☐ cubic m

23) 7 ft × 6 ft × 2 ft = ☐ cubic ft

24) 9 cm × 4 cm × 3 cm = ☐ cubic cm

Part 5 Base times height The base is already worked out.

25) 20 sq cm × 3 cm = ☐ cubic cm

26) 15 sq in × 5 in = ☐ cubic in

27) 36 sq m × 2 m = ☐ cubic m

28) 24 sq cm × 5 cm = ☐ cubic cm

29) 18 sq ft × 6 ft = ☐ cubic ft

30) 40 sq in × 2 in = ☐ cubic in

31) 12 sq m × 7 m = ☐ cubic m

32) 50 sq cm × 4 cm = ☐ cubic cm

Volume: Practice (continued)

Part 6 Find the missing edge Volume ÷ the other two edges.

33) $4 \times 5 \times \square = 60$

34) $6 \times 2 \times \square = 84$

35) $3 \times 3 \times \square = 45$

36) $8 \times 4 \times \square = 64$

37) $5 \times 6 \times \square = 120$

38) $7 \times 3 \times \square = 126$

Part 7 Two boxes No help this time. Find each volume, then add.

39) $3 \times 2 \times 2 + 4 \times 2 \times 1 = \square + \square = \square$

40) $5 \times 3 \times 2 + 2 \times 2 \times 3 = \square + \square = \square$

41) $4 \times 4 \times 3 + 6 \times 2 \times 1 = \square + \square = \square$

42) $6 \times 3 \times 2 + 3 \times 3 \times 3 = \square + \square = \square$

Volume: Practice (continued)

Part 8 Find the volume

43) 10 cm × 8 cm × 4 cm = ☐ cubic cm

44) 8 m × 4 m × 6 m = ☐ cubic m

45) 7 cm × 4 cm × 5 cm = ☐ cubic cm

46) 10 m × 5 m × 6 m = ☐ cubic m

47) 2 ft × 8 ft × 4 ft = ☐ cubic ft

48) 9 ft × 7 ft × 2 ft = ☐ cubic ft

49) 2 cm × 5 cm × 4 cm = ☐ cubic cm

50) 11 cm × 7 cm × 4 cm = ☐ cubic cm

51) 2 in × 2 in × 5 in = ☐ cubic in

52) 5 cm × 3 cm × 3 cm = ☐ cubic cm

53) 2 ft × 3 ft × 6 ft = ☐ cubic ft

54) 5 in × 4 in × 3 in = ☐ cubic in

Volume: Practice (continued)

Part 8 Find the volume (continued)

55) 11 ft × 5 ft × 5 ft = ☐ cubic ft

56) 9 ft × 3 ft × 5 ft = ☐ cubic ft

57) 3 ft × 3 ft × 2 ft = ☐ cubic ft

58) 9 ft × 6 ft × 5 ft = ☐ cubic ft

59) 3 in × 5 in × 4 in = ☐ cubic in

60) 9 ft × 4 ft × 6 ft = ☐ cubic ft

Part 9 Find the missing edge

61) 7 × 4 × ☐ = 224

62) 4 × 8 × ☐ = 160

63) 7 × 5 × ☐ = 70

64) 8 × 2 × ☐ = 32

65) 2 × 3 × ☐ = 42

66) 8 × 9 × ☐ = 144

Volume: Practice (continued)

Part 9 Find the missing edge (continued)

67) $4 \times 2 \times \square = 24$

68) $7 \times 4 \times \square = 112$

69) $2 \times 4 \times \square = 64$

70) $8 \times 4 \times \square = 288$

71) $4 \times 9 \times \square = 288$

72) $5 \times 5 \times \square = 200$

73) $9 \times 9 \times \square = 648$

74) $7 \times 7 \times \square = 147$

75) $6 \times 9 \times \square = 216$

76) $4 \times 6 \times \square = 72$

77) $6 \times 2 \times \square = 108$

78) $6 \times 8 \times \square = 96$

Volume: Practice (continued)

Part 10 Remember? Problems from earlier packets.

79) $\frac{2}{3} \times \frac{3}{5} = \frac{\square}{\square}$

80) $\frac{3}{4} \times 16 = \square$

81) $\frac{1}{2} + \frac{2}{5} = \frac{\square}{\square}$

82) $8 \times 0.45 = \square$

83) $6.5 - 2.75 = \square$

84) $38 \times 25 = \square$

Part 11 Word problems Length × width × height.

85) A fish tank is 6 ft long, 2 ft wide and 3 ft tall. What is its volume?

Number sentence _____

Answer ☐ cubic ft

Draw or show your work

86) A box holds 5 layers of cubes. Each layer has 12 cubes. How many cubes fill the box?

Number sentence _____

Answer ☐ cubes

Draw or show your work

87) A sandbox is 4 m long and 3 m wide. Sand fills it 1 m deep. How much sand is that?

Number sentence _____

Answer ☐ cubic m

Draw or show your work

Volume: Practice (continued)

Part 11 Word problems (continued)

88) A box has a volume of 80 cubic cm. Its base is 4 cm by 5 cm. How tall is it?

Number sentence _____

Answer ☐ cm

Draw or show your work

89) Challenge: Two boxes are stacked. One is 5 × 4 × 2 cm, the other 3 × 3 × 3 cm. What is the volume in all?

Number sentence _____

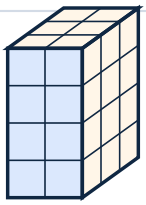
Answer ☐ cubic cm

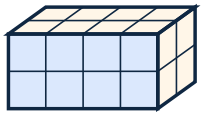
Draw or show your work

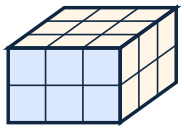
Check-In: Volume

Show what you know. No help this time. Score: _____ / 8

Part 1 Count the cubes

1)  ☐ cubes

2)  ☐ cubes

3)  ☐ cubes

Part 2 Find the volume

4) $8\text{ m} \times 2\text{ m} \times 5\text{ m} =$ ☐ cubic m

5) $5\text{ in} \times 4\text{ in} \times 4\text{ in} =$ ☐ cubic in

6) $7\text{ cm} \times 3\text{ cm} \times 2\text{ cm} =$ ☐ cubic cm

Part 3 Find the missing edge

7) $6 \times 4 \times$ ☐ $= 72$

8) $2 \times 7 \times$ ☐ $= 70$